

Intersection Geometry of Complements in a Restricted Mark Cokernel

Anonymous

Abstract

An exact restricted mark cokernel was previously split as $C = D \oplus K$, with $D \cong \mathbb{Z}/8 \oplus (\mathbb{Z}/2)^2$ and exactly 2^{41} complements. We determine the intersection geometry of the entire complement family. Relative to any fixed complement, the possible intersection indices are 1, 2, 4, 8, 16, 32, and we give the exact multiplicity of every index and every intersection-quotient type. The common intersection of all complements is $8C \cong \mathbb{Z}/3 \oplus \mathbb{Z}/18$, whereas their generated span is all of C . Finally, we compute the orders of the sixteen named classes in this common core and find exactly twenty-five generating triples, all containing the ninth class. Every claim remains on the frozen sixteen-type support.

1 Graph model and scope

Fix the C69 decomposition $C = D \oplus K$. In primary exponent notation,

$$D : (3, 1, 1), \quad K_{(2)} : (4, 2, 2, 2, 1^8), \quad K_{(3)} : (2, 1).$$

Every complement of D is uniquely the graph

$$\Gamma_f = \{(f(k), k) : k \in K\}, \quad f \in \text{Hom}(K, D).$$

The parameter group is

$$\text{Hom}(K, D) \cong (\mathbb{Z}/2)^{32} \oplus (\mathbb{Z}/4)^3 \oplus \mathbb{Z}/8, \quad |\text{Hom}(K, D)| = 2^{41}.$$

For two parameters f, g one has

$$\Gamma_f \cap \Gamma_g \cong \ker(f - g), \quad K / \ker(f - g) \cong \text{im}(f - g). \quad (1)$$

Thus all intersection statistics are translation invariant in the affine parameter group. Throughout, “intersection index” means

$$[\Gamma_f : \Gamma_f \cap \Gamma_g] = [K : \ker(f - g)] = |\text{im}(f - g)|,$$

not the index of the intersection in C .

The scope firewall is

$$\text{NO_BAD_EULER_OR_ROOT_NUMBER.}$$

We do not claim a canonical complement, a complete classification of the abstract kernel types, a full table of marks or Burnside ring, arithmetic or local data, Euler factors, root numbers, automorphy, or a Hilbert–Polya operator.

2 Exact quotient-type distribution

For a subgroup type $H \leq D$, the number of maps $K \rightarrow D$ with image of type H is

$$N_D(H) E_K(H), \quad (2)$$

where $N_D(H)$ counts subgroups of D of type H and $E_K(H)$ counts epimorphisms from K onto one fixed copy of H . The producer obtains $E_K(H)$ by strict inversion on the concrete 38-element subgroup poset of D . Independently, the checker uses

$$E_K(H) = N_K(H) |\text{Aut}(H)|,$$

because finite duality sends epimorphisms $K \rightarrow H$ to injections $H^\vee \rightarrow K^\vee$, and these groups are abstractly self-dual. Birkhoff’s subgroup formula evaluates both $N_D(H)$ and $N_K(H)$.

$H \cong K/\ker f$	$N_D(H)$	$E_K(H)$	maps with image type H
0	1	1	1
$\mathbb{Z}/2$	7	4095	28665
$\mathbb{Z}/4$	4	61440	245760
$(\mathbb{Z}/2)^2$	7	16764930	117354510
$\mathbb{Z}/8$	4	65536	262144
$\mathbb{Z}/2 \oplus \mathbb{Z}/4$	6	251535360	1509212160
$(\mathbb{Z}/2)^3$	1	68602093560	68602093560
$\mathbb{Z}/2 \oplus \mathbb{Z}/8$	6	268304384	1609826304
$(\mathbb{Z}/2)^2 \oplus \mathbb{Z}/4$	1	1029282693120	1029282693120
$\mathbb{Z}/8 \oplus (\mathbb{Z}/2)^2$	1	1097901539328	1097901539328

Table 1: Exact image, hence intersection-quotient, distribution from every fixed complement.

Theorem 1 (Intersection spectrum). *From every fixed complement, the numbers of complements at intersection indices 1, 2, 4, 8, 16, 32 are respectively*

$$1, 28665, 117600270, 70111567864, 1030892519424, 1097901539328.$$

Their sum is 2^{41} . Table 1 refines these six values into all ten possible quotient types.

Proof. Equation (1) makes the index equal to $|\text{im}(f - g)|$. Summing Equation (2) over subgroup types of each order gives the displayed values. Translation by f is a bijection of $\text{Hom}(K, D)$, so the same row occurs at every fixed complement. Multiplying by 2^{41} gives ordered-pair counts; halving the off-diagonal rows gives unordered-pair counts, all recorded in the exact evidence. $\square$

3 The common core and total span

An element common to every graph must first lie in Γ_0 , hence has the form $(0, k)$, and it belongs to all graphs precisely when every homomorphism $K \rightarrow D$ kills k . Since D has exponent eight, $8K$ is contained in this universal kernel. Conversely, the cyclic $\mathbb{Z}/8$ summand of D separates every class outside $8K$ on the dyadic primary part. The odd-primary part is killed by every map to D and multiplication by eight is invertible there. Therefore the universal kernel is exactly $8K$.

Theorem 2 (Core and span). *The complement family satisfies*

$$\bigcap_f \Gamma_f = 8C \cong \mathbb{Z}/3 \oplus \mathbb{Z}/18, \quad |8C| = 54,$$

and

$$\langle \Gamma_f : f \in \text{Hom}(K, D) \rangle = C.$$

Indeed, from every fixed complement there are 1097901539328 other parameters whose difference is surjective, and each such pair already generates C .

Proof. Because $8D = 0$, the preceding universal-kernel argument identifies the intersection with $8K = 8C$. Multiplication by eight on the primary factors of K leaves one $\mathbb{Z}/2$, together with $\mathbb{Z}/3 \oplus \mathbb{Z}/9$; combining coprime factors gives $\mathbb{Z}/3 \oplus \mathbb{Z}/18$. If $f - g$ is surjective, differences of elements of Γ_f and Γ_g generate D , while either graph maps onto K . The last row of Table 1 counts all such differences. $\square$

4 Named generators of the core

Let $e_1, \dots, e_{16}$ be the named coordinate classes in the frozen mark presentation. Exact rational residues, or independently Hermite lattices, give

$$(|8[e_j]|)_{j=1}^{16} = (9, 3, 3, 3, 1, 1, 9, 3, 2, 1, 3, 3, 1, 1, 9, 9).$$

No singleton or pair among these elements generates the order-54 core. Exactly twenty-five triples do. Every one contains $8[e_9]$; after removing that common element, the possible index pairs are

$$\begin{aligned} &(1, 3), (1, 4), (1, 7), (1, 8), (1, 11), (1, 12), (1, 15), (1, 16), \\ &(3, 7), (3, 15), (3, 16), (4, 7), (4, 15), (4, 16), \\ &(7, 8), (7, 11), (7, 12), (7, 16), (8, 15), (8, 16), \\ &(11, 15), (11, 16), (12, 15), (12, 16), (15, 16). \end{aligned}$$

This is a statement about the named presentation, not a canonical generating set for the abstract group.

5 Reproducibility and conclusion

The evidence binds the exact C64, corrected C69, and C70 bytes. Producer and checker use separate image-count routes: concrete subgroup-poset inversion versus Birkhoff counts and epimorphism duality. GAP independently enumerates all 38 target subgroups in ten types. SymPy Smith and Hermite calculations recover the common core, all named orders, and all twenty-five triples. Clean replay passes and forty-two hostile semantic mutations are rejected. C71 therefore resolves both the local pair-intersection distribution and the two global extrema of the complement family: its common core and its total span.